Anjan Dhungana

Graduate Research Assistant

Profile

Graduate student and research assistant in Poultry Science, integrating IoT, AI, and computer vision to support technology-driven poultry welfare research.

Work Experience

Graduate Research Assistant at University of Georgia, Athens, GA
August 2024 — Present

Graduate Research Assistant at Kentucky State University, KY
August 2022 — July 2024

Plant Science Instructor at Shree Sharada Secondary School, Dolakha, Nepal
May 2022 — August 2022

Agriculture Technician at DreamWork Solution, Chitwan, Nepal
February 2020 — December 2020

Education

Ph.D. in Poultry Science, University of Georgia, GA, USA
August 2024 — Current

Master of Science in Environmental Studies, Kentucky State University, KY, USA
August 2022 — July 2024

B.Sc. Agriculture, Institute of Agricultural and Animal Science, Rupandehi, Nepal
October 2015 — October 2019

Research Experience

Ph.D. Research, GA, USA

Application of Artificial Intelligence (AI) and Internet of things (IoT) to Improve Welfare Status of Laying Hens.

MES Thesis, Kentucky, USA

Effects Of *Saccharomyces cerevisiae* Fermentation Product on Weight, Methane Production, and Plasma Metabolome of Beef Steers Fed a Finishing Diet.

Undergraduate Thesis, Rupandehi, Nepal

Comparative Study on Perception and Knowledge Dynamics on Climate Change and Adaptation Measures Between Government Supported Farmers and General Farmers of Kavrepalanchok District. 2019.

Publications

Dhungana, A., Yang, X., Paneru, B., Dahal, S., Lu, G., & Chai, L. (2025). An Integrated Deep Learning Approach for Poultry Disease Detection and Classification Based on Analysis of Chicken Manure Images. *AgriEngineering*, *7*(9), 278.

Dhungana, A., Odunfa, O. A., Altman, A., Ogunade, I., McLeod, K. R., Yoon, I., ... & Jiang, Y. (2025). Effects of *Saccharomyces cerevisiae* fermentation-derived postbiotic on methane production and plasma metabolome of fattening Holstein steers fed a high-grain diet. *Journal of Animal Science*, *103*, skaf307.

Odunfa, O. A., Dhungana, A., Huang, Z., Yoon, I., & Jiang, Y. (2024). Effects of a liquid and dry *Saccharomyces cerevisiae* fermentation product feeding program on ruminal fermentation,

Details

Athens, GA, United States, +15023197752

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Research Interest

Animal Welfare
Sustainable Agriculture
Precision Livestock Management
Computer Vision
Machine Learning

Skills

Scientific Research

Machine Learning

Data Analysis

Computer Programming (Python, R, JavaScript)

MetaboAnalyst 5.0

Frontend Web Development (React.js)

Adobe Photoshop, Illustrator, InDesign

Languages

English

Nepali

total tract digestibility, and plasma metabolome of Holstein steers receiving a grain-based diet. *Journal of Animal Science, 102,* skae223.

Jiang, Y., Dhungana, A., Odunfa, O. A., McCoun, M., McGill, J., Yoon, I., & Ogunade, I. (2025). Effects of *Saccharomyces cerevisiae* fermentation product on ruminal fermentation, total tract digestibility, blood proinflammatory cytokines, and plasma metabolome of Holstein steers fed a high-grain diet. *Translational Animal Science*, *9*, txaf058.

Paneru, B., Bist, R. B., Yang, X., Dhungana, A., Dahal, S., & Chai, L. (2025). Deep Learning Methods for Automatic Identification of Male and Female Chickens in a Cage-Free Flock. *Animals*, *15*(13), 1862.

Yang, X., Lu, G., Zhang, J., Paneru, B., Dhungana, A., Dahal, S., ... & Chai, L. (2025). Tracking Poultry Drinking Behavior and Floor Eggs in Cage-Free Houses with Innovative Depth Anything Model. *Applied Sciences*, *15*(12), 6625.

Dahal, S., Yang, X., Paneru, B., Dhungana, A., & Chai, L. (2025). Automatic Detection of Foraging Hens in a Cage-Free Environment with Computer Vision Technology. *Poultry*, *4*(3), 34.

Paneru, B., Yang, X., Dhungana, A., Dahal, S., Ritz, C. W., Kim, W., ... & Chai, L. (2025). Monitoring the ramp use of cage-free laying hens with deep learning technologies. *Poultry Science*, 105858.

Conference and Meetings

International Poultry Scientific Forum (IPSF, January 2026)

Poultry Science Association Annual meeting (PSA, July 2025)

International Poultry Scientific Forum (IPSF, January 2025)

ASAS-CSAS-WSASAS Annual Meeting, Alberta, Canada (July 2024)

Kentucky Academy of Sciences Annual Meeting, Northern Kentucky University, KY, USA (November 2023)

ASAS-CSAS-WSASAS Annual Meeting, Albuquerque, NM, USA (July 2023)

Kentucky Academy of Sciences Annual Meeting, Morehead State University, KY, USA (November 2022).

Awards

Outstanding MES Student Award, Kentucky State University, KY
April 2023

Volunteering Activities

Events Committee Member

Poultry Science Graduate Student Club, University of Georgia, GA, USA
2017 — 2018

Coordinator, ICT Wing

Youth Endeavor for Sustainable Agriculture, Rupandehi, Nepal
2017 — 2018

Organizing Committee Member

Lumbini Climate Convergence, Digo Bikas Institute, Rupandehi, Nepal
2017